

Dhruv Jani

Anand, Gujarat, India

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Objective

Aspiring **Embedded Software Engineer** skilled in firmware development, PCB design, Embedded C/C++, STM32, ESP32, and real-time embedded systems. Experienced in hardware-software integration, communication protocols, and control systems for robotics and automation applications. Seeking opportunities to contribute to embedded software and electronics development while expanding expertise in intelligent and reliable system design.

Education

G H Patel College of Engineering and Technology (GCET)

B.Tech in Computer Science and Engineering (IoT) | CGPA: 9.3/10

Anand, Gujarat

Sept 2022 – May 2026

G H Patel College of Engineering and Technology (GCET)

Minor in Robotics and Automation

Anand, Gujarat

Aug 2022 – May 2026

Achievements

- Finalist – Odoo Hackathon, SVNIT (Top 50 among 1200+ teams).
- Winner – Best Business Idea, SEMCOM College; Runner-up – SSIP Project Exhibition, Gujarat.
- Twice Finalist – CVMU Hackathon for IoT and robotics projects.
- Received innovation grants from CVMU and GUJCOST for robotics research.

Experience

ISRO: Space Applications Centre (SAC)

Jan 2026 – Apr 2026

Embedded Engineer Intern – ATMG-ASA Department

- Designed and developed a 16×1 RF switch matrix operating up to 40 GHz for automated multibeam antenna measurements.
- Implemented STM32-based embedded control for cascaded SP4T GaAs MMIC RF switches using SP16T architecture.
- Developed optimized Embedded C firmware using register-level GPIO control and break-before-make switching logic.
- Performed hardware interfacing, debugging, validation, and RF switching analysis using oscilloscopes and antenna measurement setups.

EDU TANTR VENTURES PVT LTD

Aug 2025 – Dec 2025

Embedded Software Engineer Intern

- Developed an embedded simulation of an automatic washing machine controller using Embedded C/C++.
- Integrated real-time control logic, water-level detection, ISR handling, and motor actuation.
- Optimized interrupt scheduling and timing functions for accurate state sequencing and stability.

Funded Projects & Research

DD Robocon 2025 – Senior Firmware Lead

Nov 2024 – Jul 2025

Grant Rs. 6,05,000 (Charutar Vidya Mandal University)

- Led firmware development for semi-autonomous basketball robots with 3 DOF omni-drive locomotion and automated shooting.
- Implemented PID-based trajectory stabilization and synchronized multi-robot communication.

Robofest 4.0 – Underwater ROV – Firmware & Electronics Lead

Apr 2024 – Jan 2025

Grant Rs. 2,50,000 (GUJCOST)

- Engineered Kraken ROV firmware using ESP32-S3 and Raspberry Pi enabling 6-DOF underwater control.
- Developed UART communication, waterproof power circuits, and optimized thruster control systems.

DD Robocon 2024 – Junior Firmware Engineer

Nov 2023 – Jul 2024

Grant Rs. 3,97,000 (Charutar Vidya Mandal University)

- Developed firmware for autonomous and manual Mecanum drive robots with encoder feedback.
- Achieved 90%+ task completion efficiency during national-level testing and trials.

Publications

- **YOLOSense: Real-Time Waste Sorting in Healthcare Environments Using Depth-Enhanced Object Detection** – Presented at SDSTD 2025.
- **Novel Approach to Automate Fast-Multibeam Antenna Characterization in CATR** – Published at NCATEX 2026; received Best Paper Award.

Technical Skills

Languages: C, C++, Embedded C, Python, JavaScript, DSA

Embedded Systems: STM32, ESP32, Arduino, Raspberry Pi, UART, SPI, I2C, PWM, FreeRTOS

PCB & Design: Altium Designer, KiCad, PCB Layout, Circuit Design, Hardware Debugging

Frameworks & Tools: Flask, Node.js, Firebase, Git, Docker, Linux, Ubuntu, VS Code